

A Dynamic Model of Speech for the Social Sciences

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Research Question

I have a dream that one day on the red hills of
Georgia,
the sons of former slaves and ...



Research Question

How can social scientists systematically
study how things are said
—Not just what is said





Why It matters

- Speech signals much more than what is said
- Modeling of speaker's state
- Modeling of the flow of speech
→ good empirical test of theory

Approach / Method

Model: MASS — Model of Audio and Speech Structure

FIGURE 2. Illustration of Generative Model

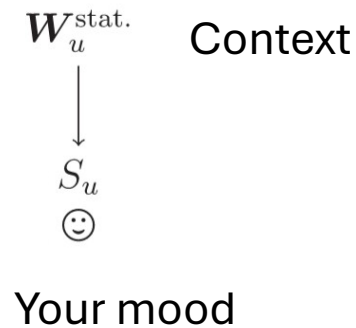
S_u
😊

Your mood

Approach / Method

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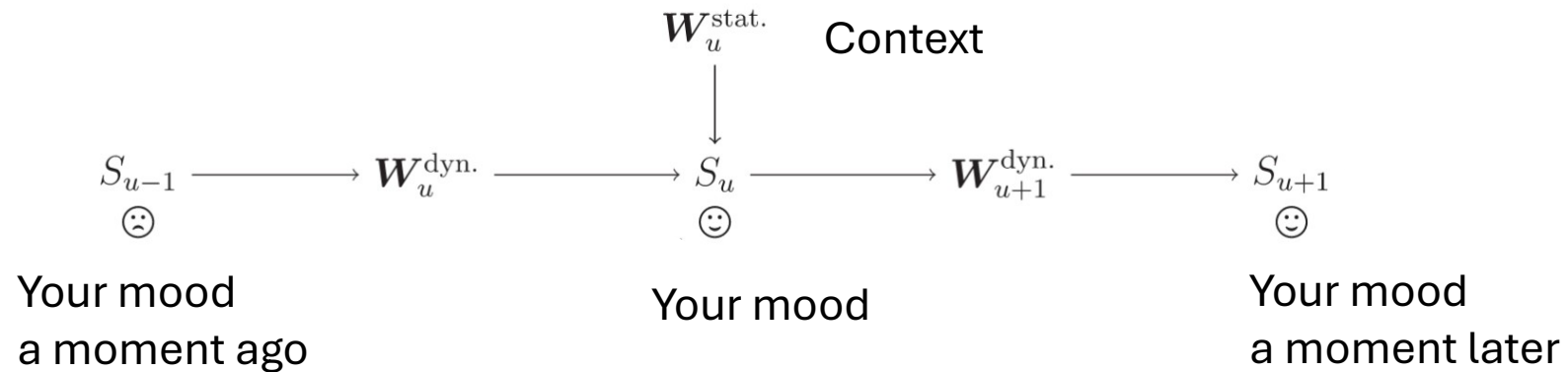
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Approach / Method

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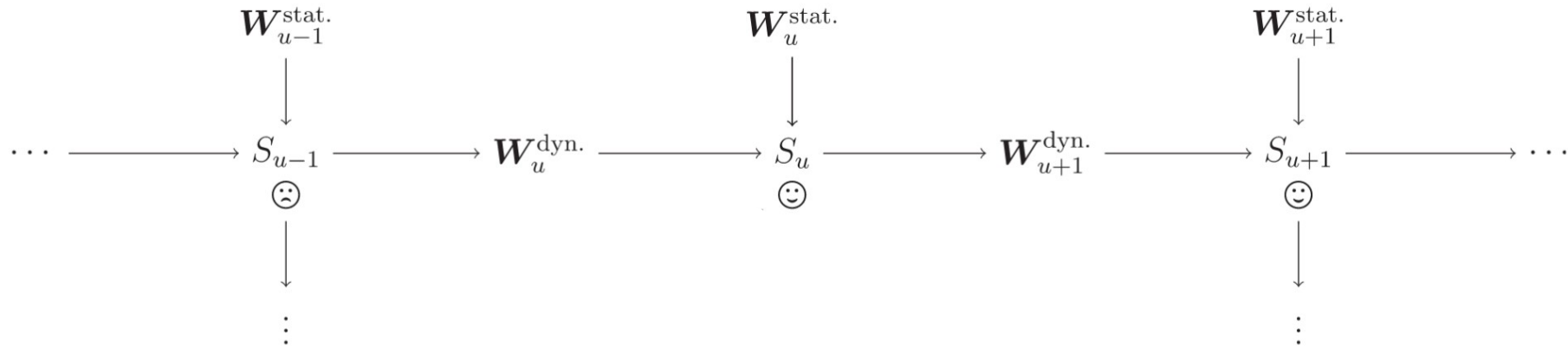
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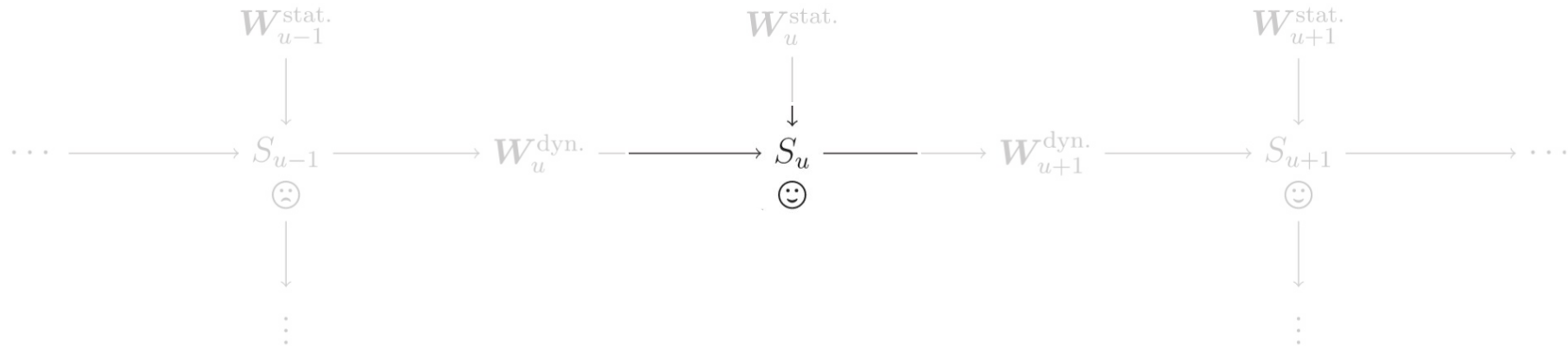


1. Multinomial logistic function

Approach / Method

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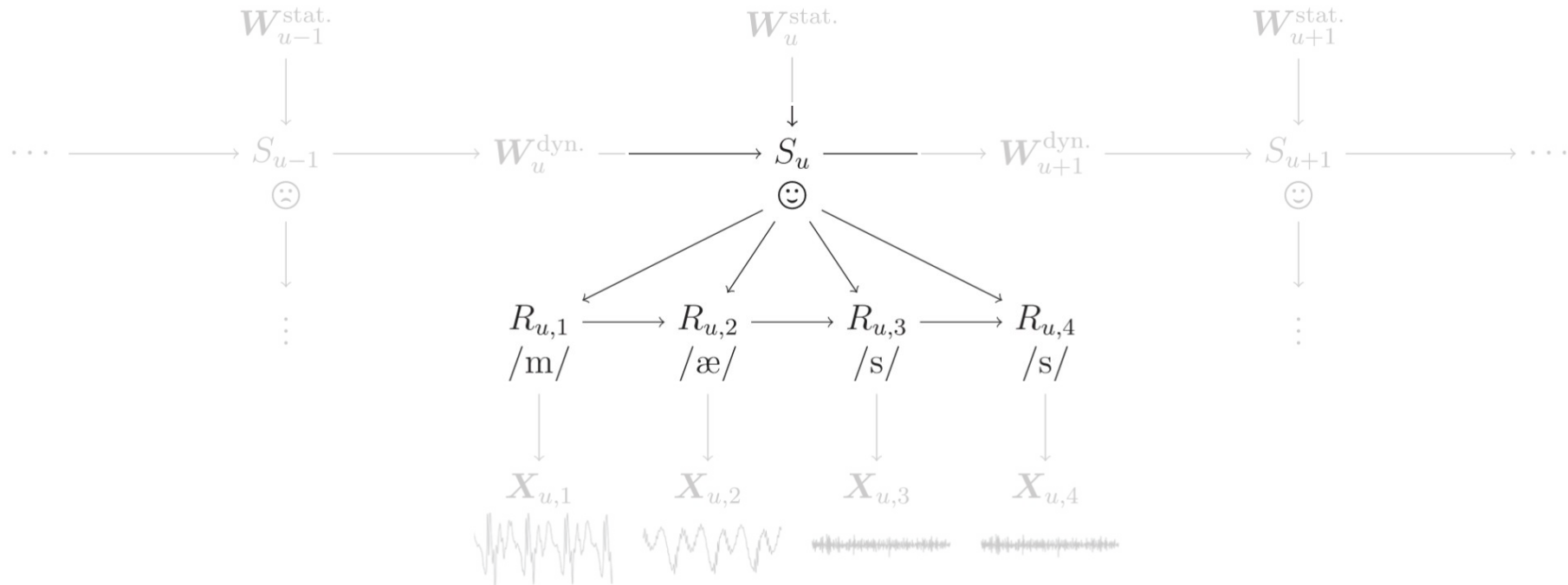
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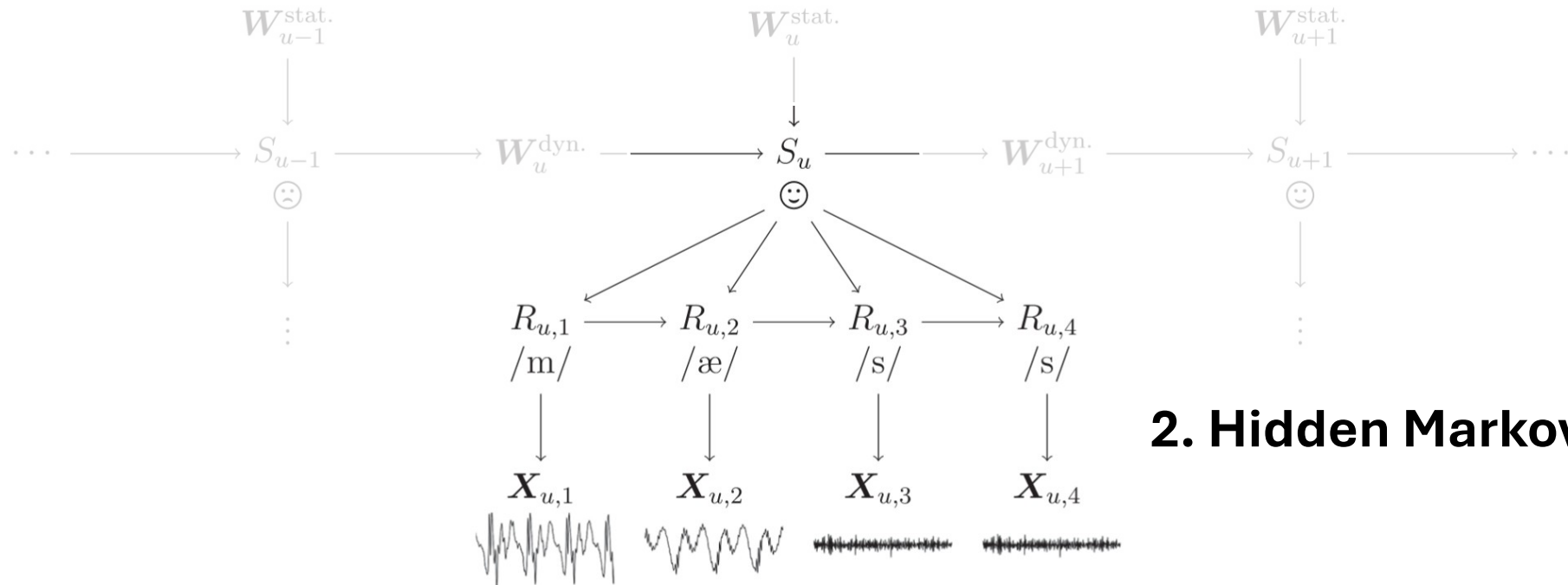
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2. Hidden Markov models

Application: U.S. Supreme Court Oral Argument

Data Summary

- 95 hours of audio, 266 arguments (2010–2016).
- Focused on 8 justices, ~42 k utterances.
- Training: 200 utterances/justice labeled as *skeptical* or *neutral*.

Estimation

- Maximize likelihood

The Quantity of interest: Judicial Skepticism

Example & Interpretation

Case: Alabama Legislative Black Caucus v. Alabama (2014)

- MASS detects when justices shift from neutral to skeptical tone.
- E.g. Scalia's sarcastic interjection → Breyer's defensive tone → Kennedy's skeptical probing.
- Skepticism is expressed through **higher loudness, greater pitch variation, vocal tension.**

Varidation & result

Out of sample performance

	Who answer	Performance
The model presented	MASS	68%
	random	52%
Human	nonexperts' majority vote	70%
	Nonexperts' individual vote	65%
Only text	Text-based classifier	50%



Demonstration

Testing theories of supreme court deliberation

justices strategically communicate
or
they are genuine fact-finding

Justices are indeed more skeptical toward the side they vote against;
but **the gap does not increase in close cases**

Suggesting genuine fact-finding style

Implications

research

- Opens the audio channel for computational social science
- Enables dynamic modeling of interaction

society

- Better understanding of institutional deliberation

Discussion questions

1. “Should governments or journalists use tone-analysis models to infer intent or sincerity in political speech?

Where is the line between analysis and surveillance?”

2. If you want to make more complicated model, do you think if assigning complicated latent variable is realistic to do supervised learning manner?